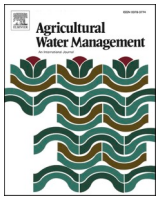
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Private service provision contributes to widespread innovation adoption among smallholder farmers: Laser land levelling technology in northwestern India

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ABSTRACT

Small farms often face challenges in adopting indivisible technologies, such as agricultural machinery, which contrast with more easily divisible technologies like new seeds and fertilisers. This study investigates key institutional factors promoting the adoption of laser land levelling (LLL), a water-saving technology that has gained wide popularity among farmers in northwestern India. The main objective is to evaluate how individual private service providers, offering LLL on a rental basis, influence technology adoption among farmers. Plot-level data from 1661 farm households across 84 villages in Punjab and western Uttar Pradesh were collected and analysed to understand farmers' technology perceptions and adoption decisions. Results from regression models show that access to a larger number of private service providers in the local context is associated with higher rates of LLL adoption, also after controlling for other farm, household, and contextual characteristics. The effect of private service provision on adoption is particularly pronounced among the smallest farms. These findings suggest that conducive institutional environments, tailored to the specific needs of different farm sizes, help to speed up innovation adoption. Our study is the first to show that individual service provision through private agents is a promising mechanism for inclusive technology adoption in the small-fam sector.

1. Introduction

Adoption of agricultural machinery is typically slow among smallholder farmers (Belton et al., 2024; Ruzzante et al., 2021). One key reason is that machinery is not easily divisible, which differentiates it from many other agricultural technologies, such as new seeds and fertilisers. Indivisible technologies are often costly and cannot be tested in small quantities for gaining more experience before fully adopting them (Lu et al., 2016). Hence, adoption rates of many indivisible technologies remain low in the small-farm sector.

One exception is laser land levelling (LLL), a water-saving technology, more formally also known as laser-assisted precision land levelling, which is widely adopted in northwestern India (Aryal et al., 2020; Villalba et al., 2024). Adoption of indivisible technologies can be facilitated through service providers that rent out machinery (Keil et al., 2021; Lu et al., 2016). The availability of rental services for LLL makes the technology accessible to farmers who cannot or do not want to own the equipment themselves. In this study, we therefore analyse how the availability of private service providers influences farmers' use of LLL technology. We hypothesize that a higher number of service providers in a farmer's proximity leads to greater adoption of LLL.

However, private service providers may not make LLL technology equally accessible to all farmers. In particular, service providers may prefer serving larger farms and larger plots to exploit economies of scale. In addition, farmers with smaller landholdings may be liquidity-constrained and risk-averse, making them less attractive business partners for private service providers (Hu et al., 2022). On the other hand, competitive and well-functioning rental markets may potentially be particularly beneficial for smallholders. Hence, we also examine whether the availability of service provision has differential effects on LLL adoption among smaller and larger farms and plots.

For our analysis, we use plot-level data from 1661 farm households

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across 84 villages in the states of Punjab and western Uttar Pradesh, located in northwestern India. We add to the literature in several ways. First, while a few studies on LLL adoption exist, all primarily focus on demand-side drivers of adoption, such as farm and farmer characteristics (e.g. farm size, soil fertility, cropping system, age, education, gender) or household characteristics (e.g. household size, off-farm income, access to credit) (Ali et al., 2018; Aryal et al., 2018, 2020; Pal et al., 2021; Sheikh et al., 2022). We are particularly interested in the role of private service providers as a potential supply-side driver of adoption. Second, much of the existing literature looks at farmers' adoption decision as a one-time choice. However, adoption is often a process that starts before the actual decision to use a technology for the first time and also continues afterwards. For example, in the case of LLL, there are at least two important decisions after the first adoption that have additional time dimensions, namely (i) the frequency of levelling (e.g. once every year or once every two or three years), and (ii) at what time in the year farmers use LLL. Such dynamics need to be understood in order to address possible adoption constraints effectively (Ishtiaque et al., 2024). We explore some of the relevant dynamics by analysing the timing of LLL adoption, farmers' perceptions of technology effects, as well as the frequency of technology use, given that land preparation and levelling decisions have to be made repeatedly.

The remainder of this article is structured as follows. In Section 2, we provide some more background on the LLL technology and how it was introduced in India. In Section 3, we discuss the theoretical framework of the technology adoption analysis, whereas in Section 4 we introduce the empirical approach. The results are presented and discussed in Section 5, while Section 6 concludes.

2. Laser land levelling technology and service providers

LLL technology was developed in the USA in the 1970s, and subsequently manufactured and disseminated in other countries, including Italy, Russia, Egypt, India, Pakistan, China, Iran, Vietnam, Cambodia, Nepal, and Tajikistan, among others (Chen et al., 2024). In India, the technology was introduced in 2001 by the International Maize and Wheat Improvement Center (CIMMYT) and the International Rice Research Institute (IRRI) along with national partners (Indian Council of Agricultural Research and State Agricultural Universities). The primary objective of introducing LLL in India was to solve the issue of rapidly declining groundwater levels (Aryal et al., 2018). In northwestern India, in particular, the over-extraction of groundwater for rice and other crops has resulted in the world's largest 'groundwater footprint', with potentially serious consequences for future agricultural production potentials (Jain et al., 2021).

Land levelling is an operation undertaken by farmers before growing a crop. It facilitates a more uniform distribution of water and fertilisers, which is essential for efficient input use and high yields (Jat et al., 2006; Chen et al., 2024). Proper land levelling is particularly important in rice-wheat systems under flood irrigation and where a certain water depth must be maintained for rice cultivation (Jat et al., 2006; Nguyen-Van-Hung et al., 2022). Unlike the traditional approach of land levelling, namely to use wooden or iron planks, LLL technology is more precise: with its precision-guided system, LLL can achieve a smoother surface (±2 cm) (Jat et al., 2006). LLL technology consists of a tractor-mounted bucket scrapper with a receiver, a control box in the tractor, and an independent transmitter on a tripod (Fig. 1).

Purchasing LLL technology is costly, which is seen as an important adoption hurdle for smallholder farmers (Larson et al., 2016). An often-used policy strategy to address accessibility issues is the establishment of a system renting out the technology through co-operatives. However, in northwestern India, LLL technology is mainly accessed through private service providers who are oftentimes farmers themselves (Aryal et al., 2018; Gulati et al., 2017). In Punjab and western Uttar Pradesh, the government offers subsidies of about 80 % and 50 % to co-operatives and individual farmers, respectively, for LLL purchases.



Fig. 1. Laser land levelling technology operated at night in northwestern India, highlighting the demand for the technology. *Note*: The technology consists of a tractor-mounted bucket scrapper with a receiver, a control box in the tractor, and an independent transmitter on a tripod. The transmitter transfers signals as a laser beam (which is why the technology is called laser land leveller) to the receiver attached to the bucket scrapper, which removes or adds soil using a hydraulic system. The tractor operator can further adjust the levels using the control box in the tractor. See Rickman (2002) for more details. For a video animation, see: https://www.youtube.com/watch?v=kRAwyr6oK7Q. *Photo Source*: Dr. Anirudh Mar (Special arrangement).

These subsidies facilitated a rapid increase in the number of LLL machinery in Punjab and western Uttar Pradesh, from less than 1000 in 2003 to more than 90,000 in 2015 (Sidhu et al., 2008; Jat et al., 2006; GoI Government of India, 2023). The strong demand for this technology has led to the designing, assembling, and local manufacturing of LLL machinery in the region (Paudel et al., 2023).

3. Theoretical background

The earliest literature explaining the adoption of indivisible technologies is the threshold model (David, 1966). It assumes that farmers can adopt a technology only through own purchase; in this model, farmers will only adopt when they exceed a certain critical level of landholding. However, Feder et al. (1985) observed that the adoption of indivisible technologies by smallholders can also happen through rental services. Sunding and Zilberman (2001) proposed the generalised threshold model, addressing limitations of the traditional threshold model. The generalised threshold model accounts for farmer heterogeneity and a dynamic adoption process. It further assumes that technology adoption through renting can be a risk-reducing strategy. Building on the generalised threshold model, Lu et al. (2016) developed a framework accounting for heterogeneity in land size and conditions in which renting of technology by service providers emerges.

In their framework, Lu et al. (2016) hypothesise that farm size and land quality influence the threshold at which the decision to own or rent a technology becomes profitable for farmers. That is, farmers with small landholdings can access the technology by renting instead of purchasing it. In our study, we test this hypothesis empirically by exploring the relationship between the availability of rental services for the technology and farmers' LLL adoption.

Over the past decade, there has been a growing body of literature on rental services for agricultural machinery, particularly focusing on rental models suitable for smallholder farmers. Mottaleb et al. (2017) examine smallholder access to agricultural machinery services and identify the important role of lead farmers, who invest in machinery and rent these services to others. They find that education, credit availability, and risk-taking attitudes are significant factors in determining whether a farmer becomes a lead farmer. Similarly, Zhang et al. (2017) note that specialized mechanization service providers in China travel across the country, offering competitive harvesting services that help smallholders remain viable. Takeshima (2017) further highlight the benefits of custom hiring tractor services in Nepal, demonstrating that renting machinery can boost smallholder productivity. Additionally, Daum et al. (2021) explore an "Uber"-like model for hiring tractors, which introduces institutional mechanisms to reduce transaction costs. Lastly, Belton et al. (2021, 2024) investigate the rapid emergence of mechanization in agriculture through outsourcing, emphasizing the evolving landscape of agricultural machinery renting services.

In the case of LLL, different types of institutions can act as rental service providers, including farmer co-operatives, custom hiring centres, or private enterprises (Daum et al., 2021; Jones-Garcia and Krishna, 2021; Mottaleb et al., 2019; Villalba et al., 2024). Previous research shows that farmers are willing to pay for land levelling operations (Lybbert et al., 2018, Paudel et al., 2023), suggesting that private service providers can play an important role in the adoption of LLL technology. However, linkages between such service provision and actual adoption decisions of farmers are, so far, poorly understood (Gulati et al., 2017; Schut et al., 2020; Van Loon et al., 2020). We expand the literature on the effect of private rental services on technology adoption and – in this connection – also explore the role of farm size.

4. Materials and methods

4.1. Study area, sampling, and data

We collected data from farmers in the regions of Punjab and western Uttar Pradesh in northwestern India. Both regions are known for their rapid depletion of groundwater resources (CGWB, 2021). The study sites encompass eight districts, including Ludhiana, Fatehgarh Sahib, Sangrur, and Patiala in Punjab, and Saharanpur, Baghpat, Shamli, and Muzaffarnagar in western Uttar Pradesh (Fig. 2). We selected these districts purposively to reflect different conditions and cropping patterns. Punjab is known for its rice-wheat cropping system, whereas most farmers in western Uttar Pradesh practice a more diversified system, including sugarcane, rice, and wheat. Most previous work on LLL adoption and impacts focuses on the rice-wheat system alone (Aryal et al., 2015; Ali et al., 2018; Gulati et al., 2017; Larson et al., 2016; Lybbert et al., 2018; Paudel et al., 2023).

The study districts in Punjab and western Uttar Pradesh share similar socioeconomic attributes, are located in the same climate zone (semi-arid temperate), and have similar soil characteristics. However, they differ in terms of irrigation policies and groundwater extraction levels. The Punjab Preservation of Subsoil Water Act of 2009 mandates delayed rice sowing in Punjab to conserve water (Tripathi et al., 2016). Such a policy is not in place in western Uttar Pradesh. Additionally, irrigation electricity tariffs and availability vary between the two states. Punjab offers free irrigation electricity in eight-hour blocks, whereas western Uttar Pradesh applies a fixed rate based on pump horsepower available during day time (Sidhu et al., 2020). The study districts have different groundwater extraction rates. In Punjab, all four districts are classified as "overexploited", whereas in western Uttar Pradesh, one is classified as "overexploited", one as "critical", one as "semi-critical", and one as "safe" (Fig. 2).

For the study, we conducted a survey of 1661 farm households in the eight districts. In Punjab, the survey was implemented from June to August 2021, and in western Uttar Pradesh from October to December 2021. In the two states and eight districts, villages and farm households were selected randomly. From the selected four districts in Punjab, we randomly sampled 52 villages and 1021 farm households proportionate to the size of each village. In western Uttar Pradesh, we randomly sampled eight villages per district and 30 households from each village (640 farm households). The data collected are representative of two major cropping systems – rice-wheat and sugarcane-rice/wheat – in northwestern India.

In each household, we carried out structured personal interviews to collect detailed data on farm and household characteristics, the adoption of LLL technology at the plot level, perceived impacts of LLL, and the availability of service providers in the village or nearby. Detailed biophysical attributes of each plot and crop cultivation data for the two most recent seasons prior to the survey were also compiled. In addition, detailed input and output data were collected from all plots under cultivation by the sample households (a total of 3369 plots).

4.2. Empirical framework

We use the farm household survey data to analyse LLL diffusion among farmers in Punjab and western Uttar Pradesh over time, as well as farmers' perceptions about the impacts of this technology on crop yields, the use of water and other inputs, and crop profits. These analyses use simple descriptive statistics.

In addition, we use regression models to examine determinants of LLL adoption with a particular focus on the role of private service providers. For this, we estimate a probit model as follows:

$$P(Y_i = 1) = \Phi(\beta_0 + \quad \beta_1 servprov_i + \theta \boldsymbol{X}_{ik} + \quad \mu_i) \tag{1}$$

where $P(Y_i = 1)$ is the probability of LLL adoption on plot i. This binary outcome variable takes the value of one if a farmer used LLL on plot i in the season prior to the survey (2020/21), and zero otherwise. Note that we also run an alternative adoption model in which the binary variable takes the value of one if LLL was used in any of the three previous seasons, given that farmers do not necessarily use LLL every year. The

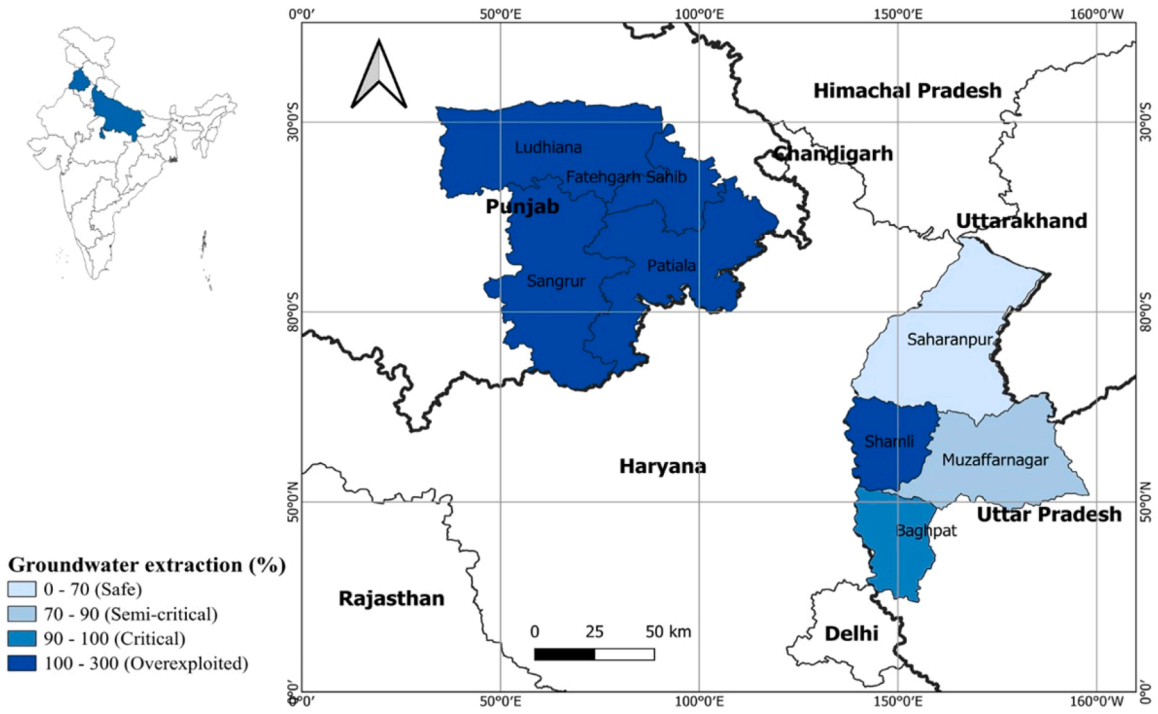


Fig. 2. Map of study area showing groundwater extraction rates at the district level. *Source*: Developed by authors based on data on groundwater extraction from the Central Ground Water Board (CGWB), Hyderabad, India.

key explanatory variable is $servprov_i$, which is the self-reported number of LLL service providers available within the village of the farmer cultivating plot i, or sufficiently nearby such that the LLL service could be accessed. The main coefficient of interest is β_1. A positive β_1 would support our first hypothesis (H1) that a larger number of service providers locally available leads to higher adoption of LLL. $\boldsymbol{X}_{ik}$ is a vector of k control variables at the plot, household, and village level that may also influence LLL adoption (see details below). $\Phi(.)$ in Eq. (1) is the probability distribution function of the standard normal distribution with independent variables as a linear combination.

Next, we are interested in understanding whether the availability of service providers has differential technology adoption effects for smaller and larger plots and farms. Specifically, we test the hypothesis (H2) that an increasing number of locally available service providers reduces possible differences in adoption between smaller and larger farms and plots. To test this hypothesis, we use the following two probit models with additional interaction terms:

$$P(Y_i = 1) = \Phi(\gamma_0 + \gamma_1 plsize_i + \gamma_2 servprov_i + \gamma_3 servprov_i \times plsize_i + \Gamma \boldsymbol{X}_{ik} + \varepsilon_i) \quad (2)$$

$$P(Y_i = 1) = \Phi(\delta_0 + \delta_1 fasize_i + \delta_2 servprov_i + \delta_3 fasize_i \times servprov_i + \Delta \boldsymbol{X}_{ik} + \varepsilon_i) \quad (3)$$

In Eq. (2), we introduce the plot size and an interaction term between the plot size and the number of service providers. A positive coefficient γ_1 would indicate that LLL adoption is more likely on larger plots. A positive interaction coefficient γ_3 would indicate that the effect of the number of LLL service providers on adoption increases with plot size (and vice versa). Eq. (3) follows the same structure but looks at farm size instead of plot size. Plot size and farm size are not the same, as most farms cultivate more than one plot.

In addition to looking at the individual coefficients of plot and farm size and the interaction terms in Eqs. (2) and (3), we also calculate the marginal effects of service providers on LLL adoption as follows:

$$\frac{\partial P(Y_i = 1)}{\partial servprov_i} = \gamma_2 \Phi(\boldsymbol{X}_i) + \gamma_3 plsize_i \Phi(\boldsymbol{X}_i) \quad (4)$$

$$\frac{\partial P(Y_i = 1)}{\partial servprov_i} = \delta_2 \Phi(\boldsymbol{X}_i) + \delta_3 fasize_i \Phi(\boldsymbol{X}_i) \quad (5)$$

These marginal effects are calculated at the mean values of the covariates $\boldsymbol{X}_i$. We show these effects graphically for different numbers of service providers. We also test for possible violations of model assumptions, including multicollinearity and endogeneity, using variance inflation factors (VIF) and the Ramsey RESET specification test.

4.3. Control variables

The control variables ($\boldsymbol{X}_i$) used in our regression models are chosen based on the existing literature on LLL adoption (Ali et al., 2018; Aryal et al., 2018, 2020; Sheikh et al., 2022). These variables, their units of measurement, and sample mean values are shown in Table 1.

Roger (2003) suggests that the spread of innovation is affected by various social factors, such as gender, caste, and class, as well as societal norms. Ali et al. (2018) find evidence supporting this idea, demonstrating associations between various socioeconomic factors and the adoption of LLL. Aryal et al. (2018) highlight that farmers with more education tend to have better access to information about new technologies, thus increasing the likelihood of adoption. The caste system, which still plays a significant role in India's social hierarchy, can either facilitate or hinder access to information, markets, and resources, thus also potentially affecting technology adoption (Krishna et al., 2019). Additionally, household wealth was shown to influence technology adoption, mostly in a positive way (Aryal et al., 2018). Such variables are also included in our regression models.

In terms of plot-level characteristics, we include soil type, slope, fertility, and waterlogging. Soil fertility and slope can influence the decision to adopt LLL and other water-conservation technologies significantly (Abdulai and Huffman, 2014; Ali et al., 2018; Aryal et al., 2018). Households facing water scarcity are also more likely to adopt

Table 1
Descriptive statistics of explanatory variables.

Variable name	Description	Mean (Std. deviation)
Village-specific variables (n = 84)		
Share of adopters	Share of LLL adopters in the village (minus the household) in the reference year (2020–21)	0.28 (0.18)
Groundwater level	Groundwater depth at the village level (meters)	27.11 (12.04)
Crop diversity – Kharif	Crop diversity in the Kharif season (Simpson index[#])	0.32 (0.15)
Crop diversity – Rabi	Crop diversity in the Rabi season (Simpson index[#])	0.39 (0.16)
Distance to district HQ	Distance from village centre to district headquarters (km)	19.04 (16.81)
Household-specific variables (n = 1661)		
Age of HH	Age of the household head	53.48 (13.34)
Education of HH	Number of years of education of the household head	7.50 (4.68)
Non-marginalised caste	The household belongs to one of the non-marginalized castes (dummy)	0.69
Majority religion	The religion of the household is a major religion in the state (dummy)	0.59
Number of plots	The total number of plots cultivated by household	2.03 (1.17)
Farm size	Area cultivated by the household (ha)	5.43 (6.98)
Total adult members in the household	Number of adult members in the household	4.48 (1.78)
Women share	Share of adult women in the total number of adults in the household	0.46 (0.14)
Non-farm employment	A household member is employed in non-farm activities (dummy)	0.29
Asset index	Asset index estimated from 20 agricultural productive items	0.00 (1.73)
Service providers in 2020/21	Number of service providers the household has access to in 2020–21	2.30 (2.18)
Discount on first use of LLL	The household received a subsidy for the first event of adoption (dummy)	0.02
Access to information from (dummy)	The household accessed information in the last 12 months (2020–21) (dummy) from the given source	
	Government extension agency	0.38
	Krishi Vigyan Kendra or KVK	0.44
	Progressive farmer	0.64
	Non-Governmental Organisation or NGO	0.15
	Farmer collective	0.39
	Input dealer	0.65
Plot-specific variables (n = 3365)		
Plot size	The size of the plot (ha)	3.12 (3.20)
Service provider distance	Distance of plots from the LLL service provider (km)	2.87 (2.25)
Soil type	Soil type in the plot (dummy)	
	Clayey	0.33
	Loamy	0.65
	Sandy	0.02
Soil erosion	The plot is affected by soil erosion (dummy)	0.06
Waterlogging	The plot is affected by waterlogging (dummy)	0.10
Soil fertility	Soil fertility status in the plots (farmer assessment; dummy)	
	Low fertile	0.04
	Medium fertile	0.35
	High fertile	0.61
Crop in Kharif	Crops grown in the plot during the Kharif season (June to October) 2021 (dummy)	
	Non-Basmati rice	0.51
	Sugarcane	0.30
	Basmati rice	0.09
	Other crops	0.09
Western Uttar Pradesh	The plot is in western Uttar Pradesh (dummy)	0.50

Note: [#]Simpson index (SI) is calculated using the formula $SI = 1 - \sum P_i^2$, where P_i is the share of crop i in the total crop area (0 means full specialization and 1 means maximum diversification).

LLL (Ali et al., 2018).

In terms of institutional factors, we consider various variables such as subsidies for first-time use of LLL and the availability of formal and informal extension services. Subsidies are measured as a binary variable, indicating whether or not any discounts are or were available for first-time users (Jones-Garcia and Krishna, 2021). Access to extension services, which offer training on various agricultural practices, is often linked to technology adoption (di Falco et al., 2011; Aryal et al., 2018). However, Ali et al. (2018) found no significant relationship between access to extension services and LLL adoption in India.

Furthermore, we include village-level characteristics, such as the local groundwater level, the distance of the village to the district headquarters, and the diversity of crops grown during the Kharif (June to October) and Rabi (November to April) seasons. Crop diversification is a widely promoted approach to manage water in northwestern India (Chakraborti et al., 2023) and could therefore influence the adoption of other water-saving innovations, such as LLL technology. As both are alternative water-saving approaches, we expect that high crop diversity may be negatively associated with LLL adoption. We include the share of LLL adopters in the village to control for possible peer effects. Finally, we include a dummy for western Uttar Pradesh to test for possible effects of state-level policy differences.

5. Results and discussion

We start by exploring farmers' awareness and adoption of LLL technology descriptively. Then, we analyse farmers' perceptions about the effects of LLL technology, before presenting and discussing the regression results on the impacts of private service provision on adoption.

5.1. Awareness of LLL technology

LLL technology has gained widespread recognition in northwestern India. We find that 93 % of the sample farmers in Punjab and 96 % in western Uttar Pradesh are aware of this technology (Table A1 in the Online Supplementary Material). This high level of awareness is largely due to various public-sector initiatives such as field demonstrations and participatory research trials conducted in the past (Jat et al., 2006; Sidhu et al., 2008). Surveys conducted 15 years ago already indicated the presence of LLL technology in many villages across northwestern India, even though technology adoption was still limited at that time (Krishna et al., 2012).

Today, LLL technology adoption is high. Among the farmers that are aware of LLL technology, around 85 % have already used it at some point. In Punjab, 4 % of the farmers knowing LLL technology own the machinery themselves and also act as private service providers. In western Uttar Pradesh, only around 1 % of the farmers reported to own LLL machinery themselves and provide services to others. Not all farmers who own LLL machinery also provide services to others. Several larger farms use the technology primarily to cultivate their own land, with limited time left for service provision to others. This is particularly

true in Uttar Pradesh, where farm sizes tend to be larger.

5.2. Adoption of LLL technology

In the survey, we asked farmers about the time in which LLL and related services became first available in their villages and the time they started using this technology themselves. Fig. 3 shows that availability and adoption follow a parallel growth trend over time in both regions, whereby adoption occurs with a certain delay. This delay is consistent with Krishna et al. (2012), who showed that LLL technology was available in many villages in the late 2000s but not yet widely adopted at that time. Early adopters already used the technology back then, but more widespread adoption only started after 2010. Based on these data, we argue that it was not the high adoption that led to the emergence of private service providers, but that – conversely – the emergence and spread of private service providers enabled the widespread adoption of LLL technology in the first place, overcoming smallholder adoption constraints.

Education programs spearheaded by the Department of Farm Power and Machinery of the Punjab Government, which began around 2007 and were then upscaled in later years, may also have played some role for wider technology adoption. These education programs targeted farmers, machinery operators, and also leaders of local co-operative societies (Sidhu et al., 2008).

In Punjab, the majority of the LLL adopters use service providers from within the same villages, accounting for around 60 % of the total sample (Table 2). In western Uttar Pradesh, service providers often come from outside the village. In both regions, most of the service providers are private enterprises, mostly farmers themselves. About 5–6 % of the adopters in both regions own LLL equipment themselves. Co-operative societies and larger custom hiring centres play some role for LLL services in other parts of India (Villalba et al., 2024), but their role in Punjab and western Uttar Pradesh is small. A likely reason is that many farmers in Punjab and western Uttar Pradesh own a tractor, so buying additional LLL equipment and also renting it or providing the service to others is easier than in other regions where very few farmers own a tractor.

Rental charges for LLL machines and services have shown a consistent increase between 2018 and 2021 (Table 2). In Punjab, the rental fee was Rs. 800 (~US$11) per hour in 2021, slightly higher than the Rs. 750 (~US$10) charged in western Uttar Pradesh. During the survey, respondents were also asked whether they had received any discounts from service providers for their first-time LLL adoption. While no public programs to subsidize LLL services were in place, 1–2 % of the sampled farmers reported having received such discounts for their first-time use. Private service providers have their own pricing strategies and may offer discounts to increase their customer base, which was shown to be an effective strategy to increase the likelihood of LLL adoption among smallholders (Lybbert et al., 2018).

Fig. 4 looks at the frequency of LLL use among sample farmers. Around 72 % of the farmers in Punjab and 74 % of the farmers in western Uttar Pradesh have used LLL at least once in their life. However, in Punjab the technology seems to be used more frequently: 63 % of the farmers had used the technology during the three years prior to the survey (2018–2021) and 37 % had used it in the last season (2020/21). These usage rates in Punjab are higher than in western Uttar Pradesh (42 % and 17 %, respectively). One reasons for the less frequent use of LLL in western Uttar Pradesh is the widespread cultivation of sugarcane. Sugarcane is kept in the field for two years, the first year and the ratoon year, meaning that a crop rotation with either wheat or rice takes at least three years to complete. Other possible reasons may relate to differential impacts or perceived impacts of LLL technology, which we analyse below.

5.3. Farmers' perceptions of technology effects

In the survey, we also asked farmers about their perceptions of technology effects, focussing on the influence of LLL on farming operations for their main Kharif season crops. These perceptions are summarized in Table 3, separately for Punjab and western Uttar Pradesh. Farmers' perceptions of LLL are quite positive, which is unsurprising given the high adoption rates and is also in line with previous research

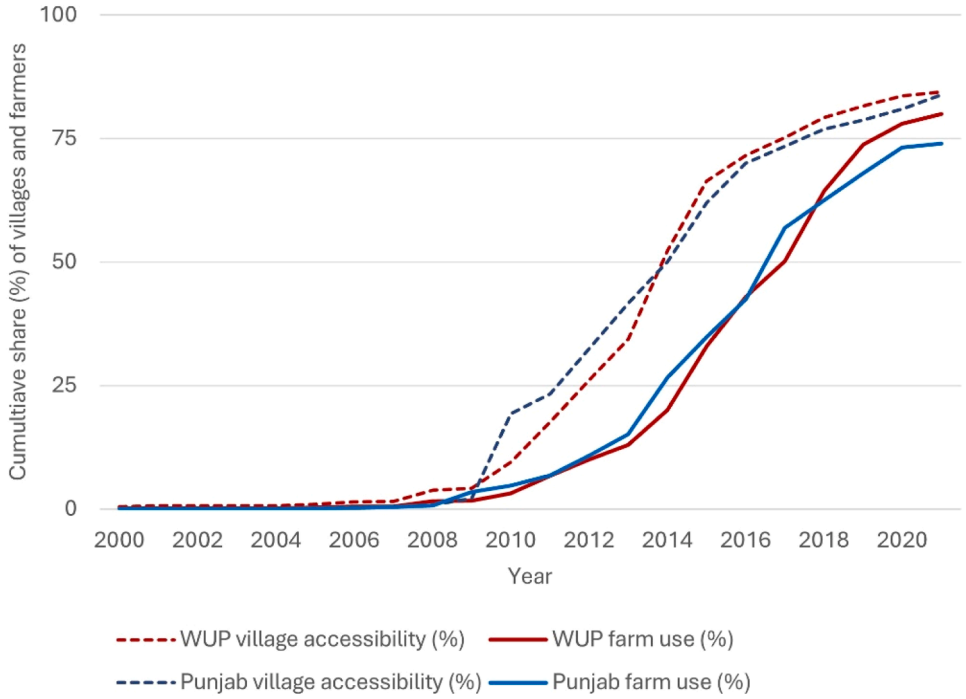


Fig. 3. Cumulative share of LLL adopters in northwestern India, 2000–2020. *Note:* We restructured the data from a cross-sectional to a panel form for the years 2000–2021 based on farmer recall, with a new dummy variable equal to one since the laser land leveller was first accessible in the village and adopted in the respondent's farm. WUP – western Uttar Pradesh.

Table 2
LLL technology trends in northwestern India (2018–2021).

LLL usage characteristics	Punjab (rice-wheat system)				Western Uttar Pradesh (sugarcane-rice/wheat system)			
	2018	2019	2020	2021	2018	2019	2020	2021
User (% of farmers)								
Before Kharif	87.15	85.49	93.07	91.88	87.79	83.44	82.89	84.81
Before Rabi	12.15	13.99	6.93	7.81	12.21	16.56	17.11	15.19
Both before Kharif and Rabi	0.69	0.50	0.00	0.31	0.00	0.00	0.00	0.00
Service provider (% of farmers)								
Own	3.82	4.15	4.82	5.63	2.91	0.00	5.26	5.06
Relative	2.08	1.04	2.11	1.25	0.00	0.00	0.00	1.27
Private / Within village	62.50	58.55	65.06	57.19	26.74	26.49	39.47	34.18
Private / Outside village	29.51	32.64	25.9	34.69	70.35	73.51	55.26	59.49
Farmer co-operative	2.08	3.11	2.11	1.25	0.00	0.00	0.00	0.00
Mean number of service providers locally available[#]	1.67	1.72	1.66	2.15	2.05	2.48	2.47	2.44
Mean rental charge (Indian Rupees in current price)	698.0	723.2	774.3	801.8	650.9	664.9	717.4	747.2

Note: Calculated based on the sub-sample of adopters (Punjab N = 755; western Uttar Pradesh N = 510). [#]Locally available service providers are service providers to which farmers have access to, either within the own village or in other villages sufficiently nearby.

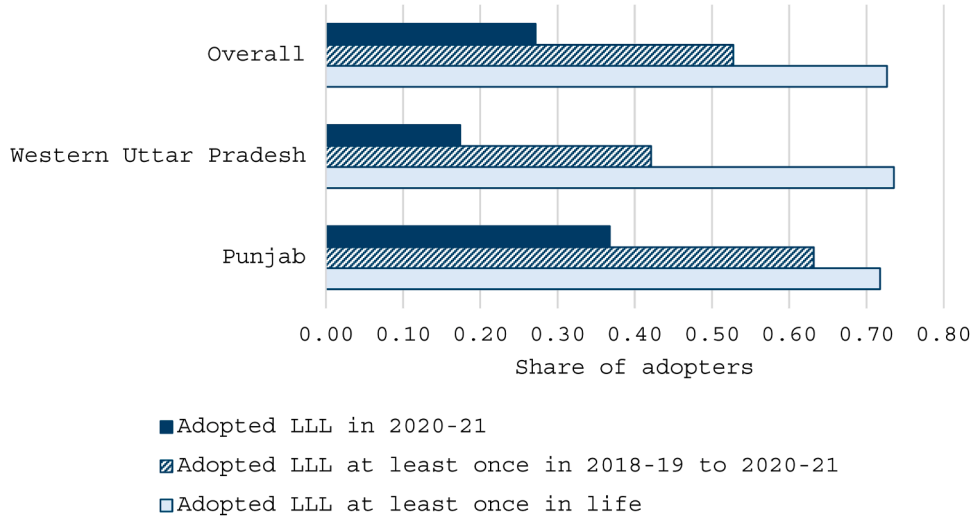


Fig. 4. Share of LLL adopters based on the frequency of technology use. *Source*: Primary data collected by authors (2021).

Table 3
Perceived impacts of LLL adoption on farming in northwestern India (share of adopters).

Perceived effects of LLL adoption on:	Punjab (n = 755)				Western Uttar Pradesh (n = 344)			
	Reduces	Increases	No change	Don't know	Reduces	Increases	No change	Don't know
Farm income	0.03	0.79	0.17	0.01	0.02	0.64	0.34	0.00
Grain yield	0.04	0.78	0.17	0.01	0.01	0.87	0.12	0.01
Cost of cultivation	0.26	0.53	0.20	0.00	0.38	0.27	0.34	0.00
Irrigation water use	0.94	0.05	0.01	0.00	0.98	0.01	0.02	0.00
Weed infestation	0.51	0.08	0.40	0.02	0.57	0.01	0.42	0.00
Burning of Kharif crop residue	0.32	0.31	0.35	0.02	0.28	0.01	0.70	0.02
Land use intensity	0.21	0.47	0.28	0.04	0.01	0.22	0.76	0.01

(Dessart et al., 2019). In both states, most farmers consider LLL to increase yields and income, and almost all farmers believe that the technology reduces the quantity of irrigation water use. These views are largely consistent with available impact research, suggesting that LLL can increase yields by about 5 % and reduce irrigation water usage by 25 % (Ali et al., 2018; Aryal et al., 2020; Larson et al., 2016; Lybbert et al., 2013; Pal et al., 2021; Sheikh et al., 2022). Field-trial results suggest that LLL-related yield gains in rice and wheat can even be higher (Jat et al., 2015).

Despite the positive overall perceptions of LLL in both Punjab and western Uttar Pradesh, some differences between the states can also be observed in Table 3. For instance, a greater proportion of farmers in Punjab (79 %) compared to western Uttar Pradesh (64 %) reported perceived increases in farm income due to LLL. In contrast, a higher percentage of farmers in western Uttar Pradesh (87 %) than in Punjab (78 %) reported perceived grain yield increases. These differences suggest that there may be regional disparities in terms of technology effects. Note, however, that in our survey questions we did not ask for the magnitude of any perceived effects. The numbers in Table 3 should not be misunderstood as actually observed or measured impacts of LLL adoption, but simply as indicators of farmers' subjective perceptions of this technology.

5.4. The role of service providers

We now present and discuss the regression results, with a particular focus on how the availability of service providers influences farmers' LLL adoption. Results from the probit model explained in Eq. (1) above are summarized in Table 4, column (1). The number of LLL service providers in or nearby the individual farmer's village is positively associated with LLL adoption in 2020/21, even though the coefficient is not statistically significant. The marginal effects for different numbers of service providers are shown in Fig. 5a. One can observe a slight increase in predicted adoption probability with an increasing number of service providers, yet with relatively large confidence intervals. These patterns suggest that the effects of service providers may be heterogenous, which we will further explore below.

Relevant control variables were included in estimation with the full results shown in Table A2 in the Online Supplementary Material (including model specification tests). Various socioeconomic variables are positively and significantly associated with LLL adoption, including involvement in non-farm employment and wealth (asset ownership). We also find that discounts on first-time use increase the likelihood of LLL adoption. Furthermore, a few village-level variables are positively associated with individual LLL adoption, namely the proportion of LLL adopters in the village and proximity to the district centre. These results are plausible and consistent with earlier research on LLL adoption in India (Aryal et al., 2015; Ali et al., 2018; Lybbert et al., 2018; Pal et al., 2021; Villalba et al., 2024). Most of the plot characteristics (e.g. soil type, soil fertility) and farmer characteristics (e.g. age, education) are not statistically significant. State-level policy differences do not seem to influence LLL adoption, as indicated by the insignificant coefficient for western Uttar Pradesh. Nor does village-level crop diversification seem to affect LLL adoption.

5.5. Heterogenous effects of service providers

As explained, we are also interested in understanding whether the local availability of service providers has differential effects on LLL adoption depending on plot and farm size. The results from the probit models explained in Eqs. (2) and (3) are summarized in Table 4, columns (2) and (3). In model (2), we include plot size and an interaction term between plot size and the number of locally available service providers. Both variables are not significant individually, but they are jointly significant with the number of service providers.

In model (2), the effect of the number of service providers itself is also statistically significant and larger than in model (1), suggesting that controlling for plot size and interaction effects, the number of locally available service providers influences LLL adoption positively. Further, the negative interaction term coefficient suggests that the positive adoption effect of service provision decreases with increasing plot size, or, in other words, the service provision effect is larger on small plots than on large plots.

To visualize these estimates, we plot the marginal effects of service provision on adoption for different plot sizes in Fig. 5b. The number of service providers has a larger positive effect on LLL adoption on smaller plots than on larger plots. In other words, the proliferation of service providers in the local contexts makes the technology more accessible to farmers with small plots. These results support our hypotheses H1 and H2.

Model (3) in Table 4 (column 3) and Fig. 5c show alternative estimates, where we look at farm size (area cultivated) rather than plot size. The effects are consistent with those of model (2). The number of service providers is positively and significantly associated with LLL adoption, but the negative interaction term coefficient suggests that this effect is primarily observed among smaller farms. Notably, farm size as such has a significantly negative association with LLL adoption, meaning that larger farms are somewhat less likely to adopt LLL technology than smaller farms. This negative association may be related to larger farms already having higher yields and easier access to irrigation water, which would lower the marginal benefits of LLL and thus decrease their incentives to adopt. However, a more detailed analysis of the impacts of LLL on small and large farms is beyond the scope of this study and would deserve further scrutiny in follow-up research.

Table 4
Probit model on determinants of LLL adoption (2020/21).

	(1) Model 1	(2) Model 2	(3) Model 3
Service providers in 2020/21	0.021	0.040$^{**\#}$	0.032$^{*\#\#\#}$
	(0.014)	(0.019)	(0.017)
Plot size	−0.015	2.E−04$^{\#}$	
	(0.010)	(0.014)	
Plot size x Number of service providers (interaction)		−0.007$^{\#}$	
		(0.005)	
Farm size			−0.013$^{**\#\#\#}$
			(0.005)
Farm size x Number of service providers (interaction)			−0.003$^{\#\#\#}$
			(0.002)
Household-level controls	Yes	Yes	Yes
Plot-level controls	Yes	Yes	Yes
Village-level controls	Yes	Yes	Yes
Model intercept	−1.160***	−1.179***	−1.355***
	(0.373)	(0.374)	(0.375)
LR Chi^2	393.18***	395.37***	403.98***
Observations$^{\$}$	2815	2815	2815

Note: *** shows significance at 1 %, ** shows significance at 5 %, and * shows significance at 10 %. $^{\#\#\#}$ shows joint significance at 1 %, and $^{\#}$ shows joint significance at 10 %. $^{\$}$The analysis is based on plot-level data from Punjab and western Uttar Pradesh, excluding households owning LLL machinery themselves (124 plots). In western Uttar Pradesh, we dropped plots on which sugarcane ratoon crop was grown in 2020/21 because levelling cannot be done before the sugarcane ratoon crop (436 plots). Full model results are provided in Table A2 in the Online Supplementary Material.

5.6. Alternative definition of LLL adoption

The analysis in Table 4 captures the determinants of LLL adoption in 2020/21, corresponding to the last season prior to the survey. However, even adopters do not necessarily use LLL technology in every season. The frequency of LLL use depends on several local agroecological factors (Nguyen-Van-Hung et al., 2022). In Table 5, we estimate the same probit models but now the adoption variable is redefined such that it considers LLL use in any of the three years prior to the survey (2018/19–2020/21). This approach helps capture adoption accounting for the sugarcane-rice/wheat crop-rotation in Uttar Pradesh. The findings in Table 5 are similar to those in Table 4. Plot size as such has a significantly positive association with LLL adoption, suggesting that on very small plots the cost of renting LLL services may be too high. But again, after controlling for plot and farm size, the number of service providers has positive effects on LLL adoption, especially among the smaller farms and those with smaller plots. These results underscore the importance of customizing agricultural support services to the specific needs of different farm sizes.

The full results of the models summarized in Table 5 are shown in Table A3 in the Online Supplementary Material. These results are also similar to those discussed above, even though a few additional control variables are now statistically significant. For instance, the farmers' age is negatively associated with LLL adoption, whereas education seems to affect adoption positively. Furthermore, the village groundwater level is significantly associated with LLL adoption: a larger groundwater depth increases the likelihood of adoption, which is plausible for a water-saving technology.

6. Summary and conclusion

In this article, we have used LLL as an example to better understand

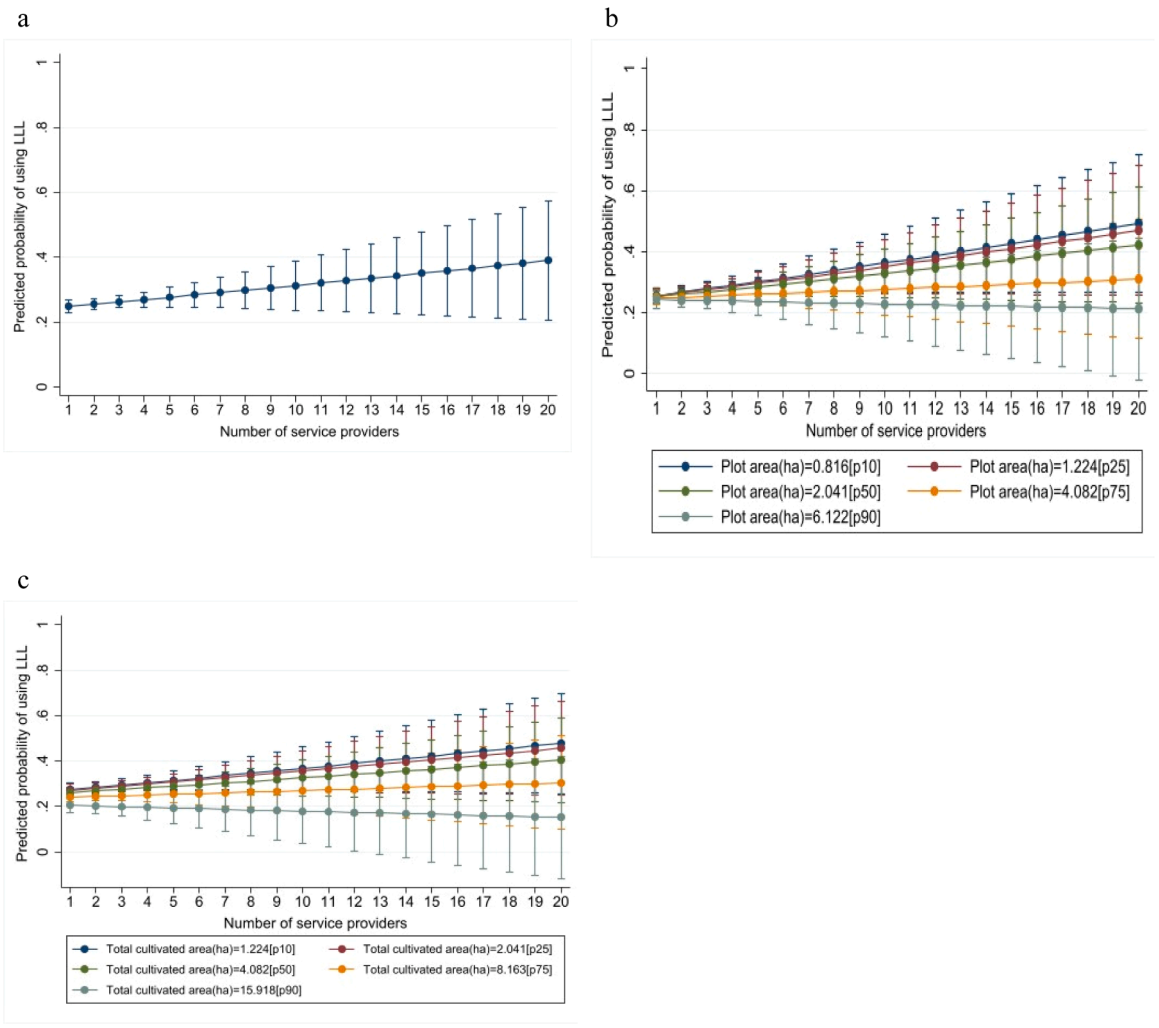


Fig. 5. Marginal effects of the number of service providers on LLL adoption. *Note*: The figure shows marginal effects calculated at sample mean values and adjusted predictions with 95 % confidence intervals. The vertical axis shows the predicted probability of LLL adoption. In panels 5b and 5c, predictions are shown for the 10th (p10), 25th(p25), 50th(p50), 75th(p75) and 90th(p90) percentile values of plot size and farm size (total cultivated area), respectively. *Source*: Estimated from regression models (1) to (3) in Table 4.

how private service providers can facilitate inclusive dissemination of indivisible technologies among smallholder farmers with fragmented plots. We have analysed how improved access to LLL renting services, measured by the number of service providers accessible either in the village or nearby, influences individual technology adoption and use. We hypothesized that a larger number of service providers would lead to more adoption, and that this effect would differ between smaller and larger farms and plots. Our results confirm both hypotheses. Our regression estimates show that the number of service providers is positively associated with the likelihood of LLL adoption, and that the marginal effect of service providers is larger on small farms and plots than on large farms and plots. In other words, small farms and plots benefit over-proportionally from better access to LLL service provision. Important to note is that the service providers in northwestern India are predominantly private enterprises, mostly farmers themselves, and are different from custom hiring centres and co-operatives.

This is the first study to show the role of individual service providers as an institutional mechanism for the adoption of indivisible technology. Our findings present a compelling case for re-evaluating traditional agricultural technology scaling models to include individual service providers for broader and more inclusive adoption. From a policy perspective, policies that promote transparent service provision in competitive rental markets can, therefore, help foster smallholder-inclusive technological change.

Our findings suggest that individual service providers are a promising mechanism for scaling LLL technology adoption in an inclusive manner. A shift from the traditional, centralized model to a service-based, flexible, on-demand model can improve the access and affordability of technology for smallholders. In the case of northwestern India, this has worked well, as can be seen in the high LLL adoption rates among smallholders. Important preconditions are available road infrastructure, to reach the plots with the machinery, information and communication technology infrastructure, to facilitate access to information and reduce transaction costs, and a critical number of farmers able and willing to invest in the machinery, to guarantee competition in the emerging rental markets. Subsidies for purchasing the equipment can help, as the Indian example also demonstrates. However, in other countries and contexts, not all of these conditions may hold, meaning that more specific public support and infrastructure investments may be needed.

More generally, our results suggest that an institutional environment that accommodates the specific needs of different types of farms can enhance broad-based innovation in the small farm sector, thus contributing to sustainable productivity growth and environmental efficiency.

A few limitations of our study should be mentioned. First, our

Table 5
Probit model on determinants of LLL adoption in at least one of the previous three years (2018/19–2020/21).

	(1) Model 1	(2) Model 2	(3) Model 3
Service providers in 2020/21	−0.003	0.013$^{\#}$	0.024$^{\#\#\#}$
	(0.012)	(0.016)	(0.015)
Plot size	0.018**	0.030$^{***\#}$	
	(0.008)	(0.011)	
Plot size x Number of service providers (interaction)		−0.006$^{\#}$	
		(0.004)	
Farm size			0.010$^{***\#\#\#}$
			(0.004)
Farm size x Number of service providers (interaction)			−0.005$^{***\#\#\#}$
			(0.002)
Household-level controls	Yes	Yes	Yes
Plot-level controls	Yes	Yes	Yes
Village-level controls	Yes	Yes	Yes
LR Chi^2	521.34***	523.73***	527.66***
Observations$^{\$}$	3237	3237	3237

Note: *** shows significance at 1 %, ** shows significance at 5 %, and * shows significance at 10 %. $^{\#\#\#}$ shows joint significance at 1 %, and $^{\#}$ shows joint significance at 10 %. $^{\$}$The analysis is based on plot-level data from Punjab and western Uttar Pradesh, excluding households owning LLL machinery themselves (124 plots). Full model results are provided in Table A3 in the Online Supplementary Material.

regression estimates show associations between the number of service providers and LLL adoption, which should not be interpreted as fully-identified causal effects. Second, the results from northwestern India may not simply be generalized to other countries and regions. In Punjab and western Uttar Pradesh, many farmers own a tractor, which facilitates the purchase of LLL equipment and the emergence of competitive rental markets. The ramifications may be different in settings where most farmers do not own a tractor. Third, while LLL is an indivisible technology, its characteristics may be peculiar. For instance, LLL is typically not used by farmers every year, so farmers who own the machinery are particularly interested to also rent it out to others for more efficient use.

Follow-up research with data from other regions and referring to other types of technologies may be useful to further add to our understanding of how the adoption of indivisible technologies in the small-farm sector can be promoted through suitable institutional mechanisms. In particular, future research should analyse the effectiveness of private service provision in situations with less-developed infrastructure and in comparison to alternative service provision models. The types of crops grown and farmers' access to markets and information may also matter. Findings from diverse settings could help to better understand the complexities of technology adoption, particularly regarding the heterogeneity of farming practices and institutional conditions.

CRediT authorship contribution statement

Qaim Matin: Writing – review & editing, Supervision. **Krishna Vijesh V:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Surendran Padmaja Subash:** Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Parlasca Martin C:** Writing – review & editing, Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.agwat.2025.109411.

Data availability

Data will be made available on request.

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